Magnetometer Construction Notes

Mark Braunstein WA4KFZ

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Revision: 1.0

Document Revisions

Date	Revision	Author	Notes
27 June 2026	1.0	Mark Braunstein WA4KFZ	Initial release

Contents

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Caution

The RM3100 breakout board contains three magnetic coil assemblies (Figure 1) that are very fragile. During construction, be sure to carefully handle the remote PC board and avoid inadvertently impacting the coils with tools or other objects.

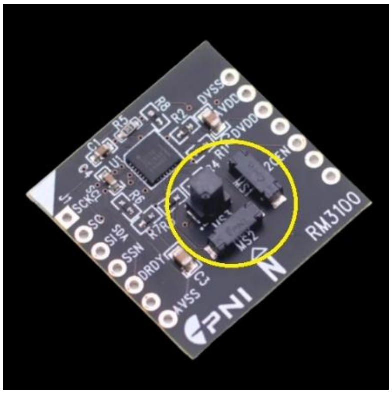


Figure 1 – Location of the three magnetic coil assemblies on the RM3100 breakout board. Be sure to avoid damaging the coils during magnetometer assembly.

Tools

- 4” bench vise or similar
- Hacksaw or miter saw (to cut PVC pipe sections)
- 3/8” or larger variable speed drill
- 1-3/8” Forstner drill bit
- 1/16” twist drill bit
- Palm sander
- 100-grit sandpaper
- 220-grit sandpaper
- Straight edge ruler
- Fine point permanent marker pen
- Long nose pliers
- Scissors or utility knife

PVC Plug Surface Preparation

Two PVC SPG plugs are used in the construction of the magnetometer housing: one on top to mount a bubble level indicator and one on the side to mount an RJ45 bulkhead adapter.

The “flat” surfaces on these plugs are unfortunately not flat enough for our applications and need to be modified.

- Place the 2” PVC SPG plugs in a bench vise with the flat side up.
- Using a palm sander with 100-grit sandpaper, press firmly to smooth the top surface of the plug. Move the sander in a circular motion. When sufficiently smoothed, there should be no “dips” in the surface. The sanded PVC will have a dull, white finish.
- Using 220-grit sandpaper, sand the top surface of the plug by hand. This further flattens the surface and improves hardware retention for the bubble level and RJ45 bulkhead adapter.

Bubble Level Attachment

A 50 x 10mm bubble level is adhered to the top PVC plug. During magnetometer installation in the ground, this bubble level aids in the Z-axis alignment process. The bubble level is attached using the provided double-sided tape.

To center the bubble level on the plug’s top surface, use a marker pen to draw lines from one octagonal edge to the opposite edge (Figure 2). The line will pass through the center of the plug. Repeating the process will define the center of the plug as the intersection of the lines.

When placing the bubble level on the flat surface of the plug, use the drawn lines to align with those of the bubble level (Figure 3). This will aid in magnetometer orientation during ground installation.

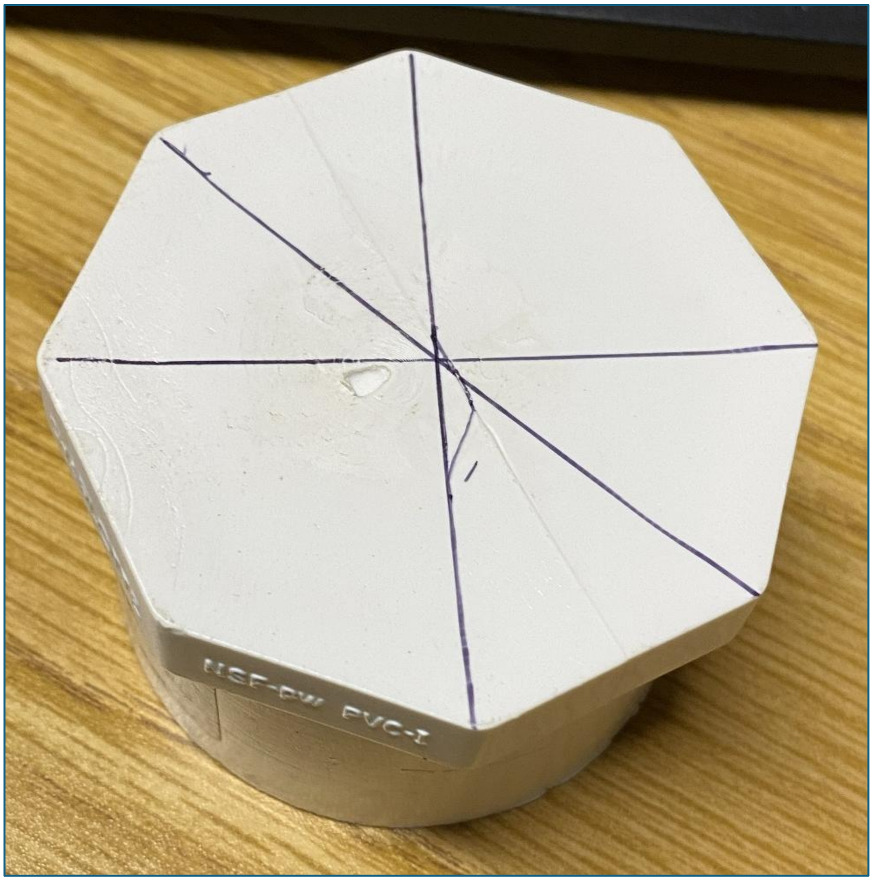


Figure 2 – A simple method to locate the center of the plug.

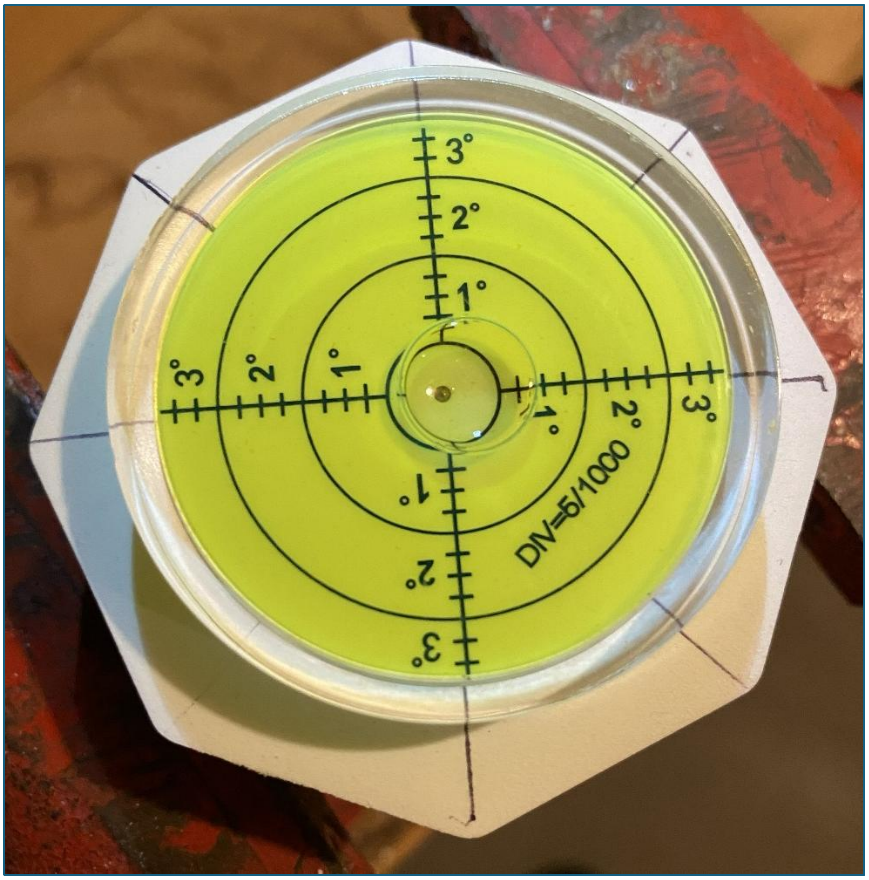


Figure 3 – Bubble level installed on the PVC plug and oriented with the drawn lines on the top surface.

RJ45 Bulkhead Adapter Attachment

A 1" hole is drilled in a PVC SPG plug to accommodate insertion of the RJ45 bulkhead adapter. The adapter barely has enough thread length to pass through the drilled PVC SPG plug to allow installation of the coupling hex nut. To provide additional thread length, it is necessary to countersink the inside surface of the plug. A 1-3/8" Forstner bit (commonly used for woodworking applications) is used to mill the inside surface. The milling depth required is approximately the height of coupling hex nut. The Forstner bit allows the milled hole to retain a flat surface.

- Drill a 1/16" hole through the center of the plug. This small hole serves to align the Forstner bit during operation.
- With the Forstner bit installed, run the drill at high speed and gently press the bit into the plug. The Forstner bit has sharp edges that can "catch" the PVC plastic and cause "bucking" of the drill. Use slow, gentle pressure to run the bit against the PVC plastic. Keep the bit as perpendicular to the plug as possible. With sufficient downward pressure, plastic chips will be "shaved" from the inside surface of the

plug. Periodically stop the drill and check the depth of the milled area. It may be necessary to slightly tilt the Forstner bit to keep the milled area perpendicular to the centerline of the plug. Continue milling until the depth of the cut equals the height of the coupling hex nut (Figure 4).

Figure 4 – Interior view of the PVC plug after drilling with the Forstner bit. Note the 1/16" hole to aid in keeping the bit centered during the drilling process. The recessed hole depth is approximately the thickness of the RJ45 adapter hex nut.

- Remove the Forstner bit and install the stepped drill bit.

- Turn the plug over and secure it in the bench vise. Use the stepped drill bit to drill a 1” hole, starting on the top surface of the plug (Figure 5).

Figure 5 – Drilling the 1” hole on the top surface of the plug.

- Turn the plug over again and repeat the drilling process from the inside of plug. This is necessary to ensure that the 1” diameter hole is uniform through the plug’s face.
- Insert the RJ45 bulkhead adapter into the modified plug. Install the coupling hex nut on the inside of the plug. Finger-tighten the nut, then use a long pair of needle nose pliers to span the hex nut and gently tighten the connection (Figure 6). The rubber gasket on the adapter (located on the top surface of the plug) provides the necessary weatherproofing for the installation (Figure 7).

Figure 6 – RJ45 bulkhead adapter installed using the hex coupling nut. Finger-tighten the hex nut first, then use needle nose pliers to tighten the nut fully.

Figure 7 – RJ45 bulkhead adapter installed on the PVC plug and ready for cable attachment.

Housing Construction

The magnetometer housing (Figure 8) is constructed from Schedule 40 PVC pipe (commonly found at hardware stores) and associated PVC fittings. The pipe sections can be cut to length using a hacksaw or miter saw. Construction of the housing is shown in the figure. Use PVC solvent cement to join the indicated fittings to the pipe sections. The PVC SPG plugs are pressed into the tee-fitting by hand and weather sealed with wrapped tape or silicone caulking. If necessary, a large wrench can be used to loosen these plugs for removal.

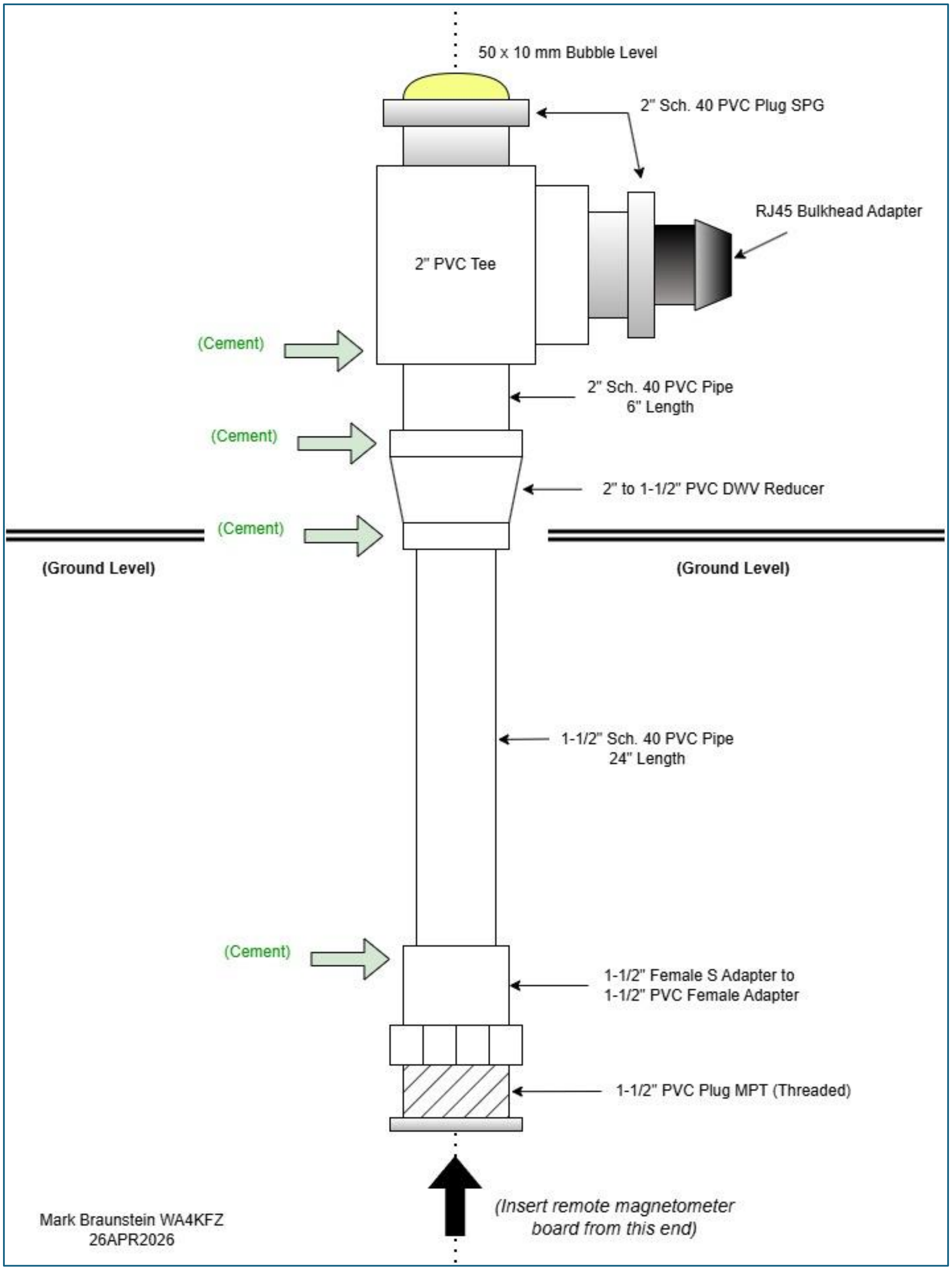


Figure 8 – Magnetometer housing construction using PVC plumbing pipe and fittings.

Remote PC Board Installation

Before installing the remote PC board inside the housing, feed the 3ft CAT6 patch cable through the 2” PVC Tee fitting (Figure 9). The cable will then protrude through the bottom of the 1-1/2” pipe end.

Figure 9 - Inserting the patch cord through the PVC Tee fitting.

On the connector that plugs into the RJ45 bulkhead adapter, it was necessary to cut off the protective rubber boot around the connector release tab (Figure 10). A utility knife can easily cut through the material.

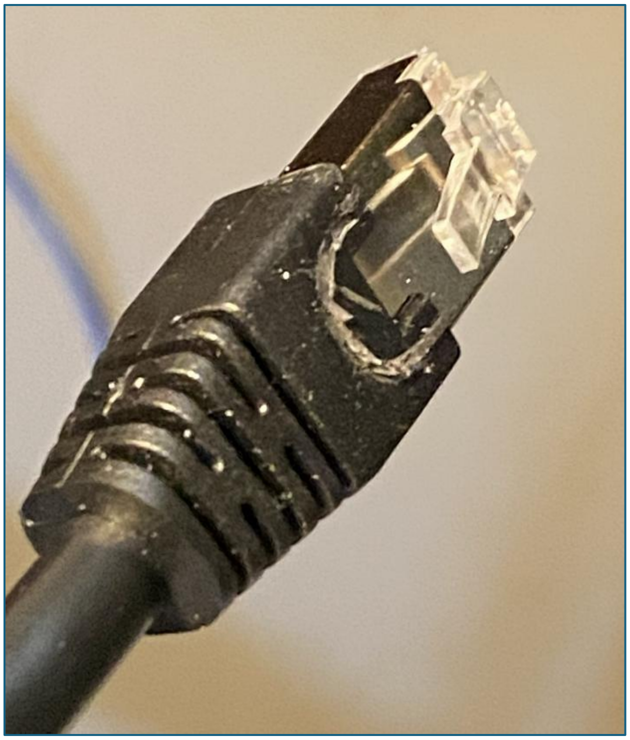


Figure 10 - The portion of the rubber boot that protects the connector release lever was cut off using a utility knife.

Plug one end of the cable into the remote PC board connector. The remote PC board is installed in the magnetometer housing **from the bottom end**. Since the board's width is less than the interior diameter of the 1-1/2" PVC pipe, it is necessary to "shim" the board to keep it centered and secured. If a 3D printed fixture is not available, a 4" x 4" piece of 4mm-thick closed cell foam can be used to wrap around the board (Figure 11) and create a friction fit to hold the board in place (Figure 12).

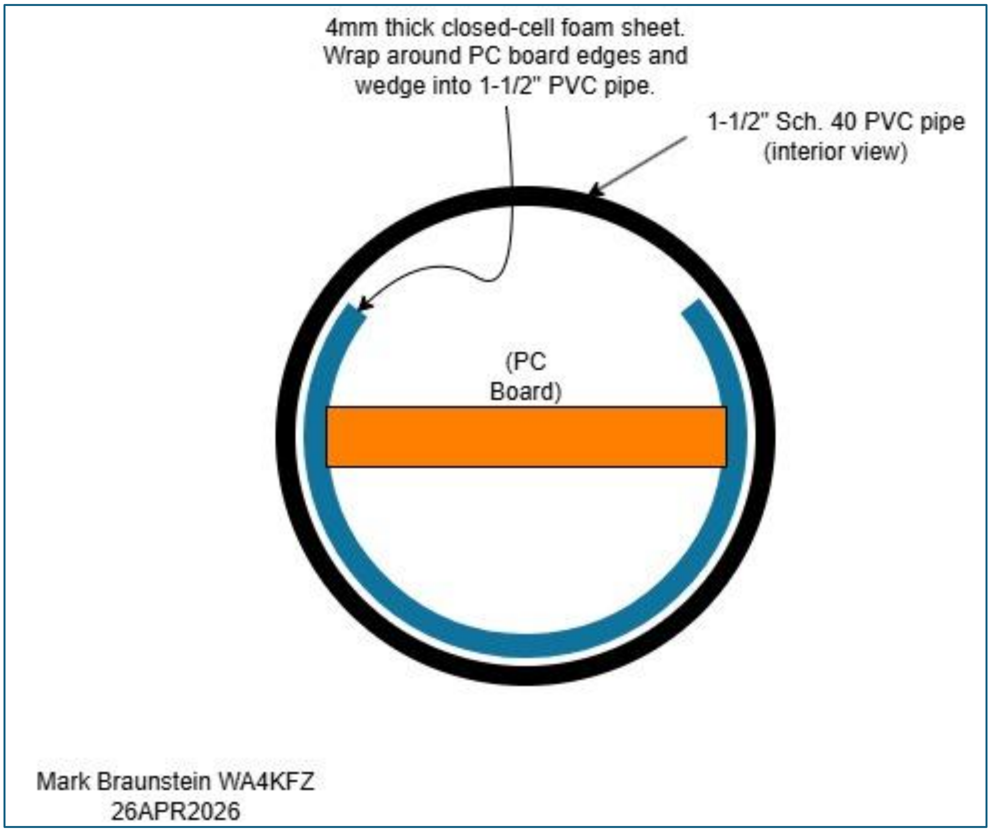


Figure 11 - Wrap 4mm-thick closed cell foam material around the PC board and gently push the assembly into the bottom end of the 1-1/2" PVC pipe.

Figure 12 - The remote PC board installed at the bottom end of the 1-1/2" PVC pipe. The PC board is held in place using 4mm-thick closed cell foam.

When inserting the PC board and foam material, gently push the combination into the end of the pipe. If it becomes necessary to remove the PC board, use long nose pliers and **grab the foam material** while aiding removal with your hand. **Do not use the pliers on the PC board itself; you'll risk damaging the board.**

The PC board is oriented so that the tall sensor coil faces in the direction defined as "North" during the ground installation step (Figure 13).

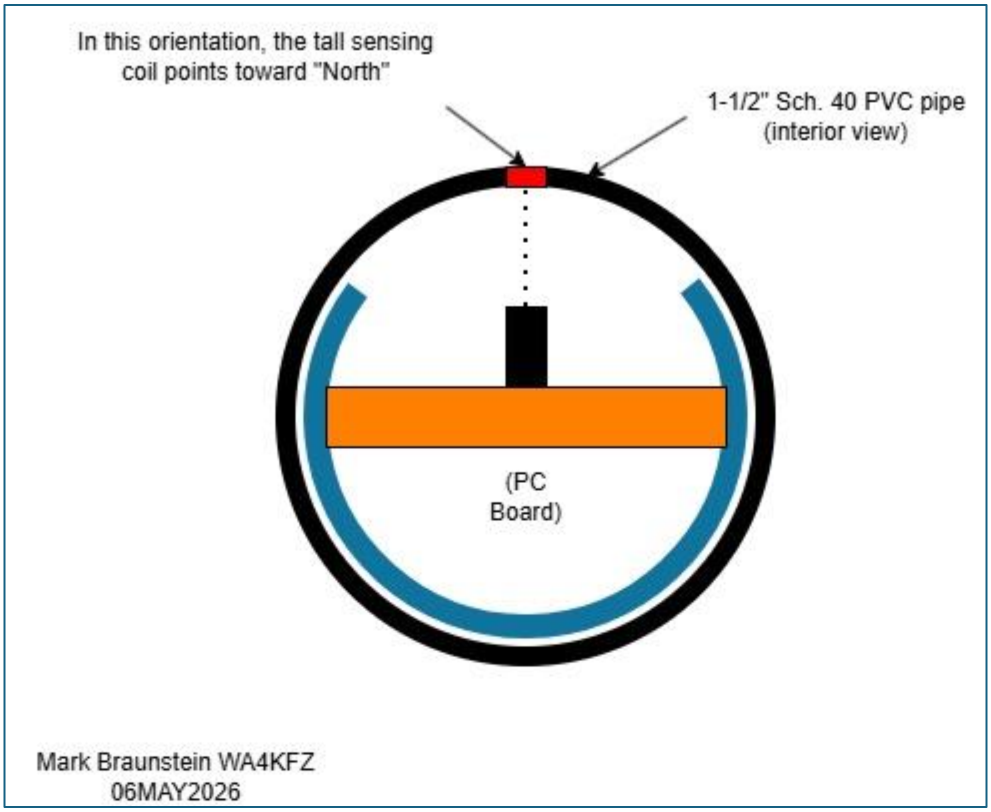


Figure 13 - Orientation of PC board inside the PVC pipe. The tall sensing coil points toward "North."

Once the PC board has been installed in the housing, plug the other end of the cable into the RJ45 bulkhead adapter. Firmly press the PVC plug with the adapter into the 2" PVC Tee.

In the final step, the bottom threaded plug is inserted once the PC board has been installed. Using a large wrench, tighten the plug into the fitting until it stops rotating. Do not overtighten the plug.

Environmental Sealing

Most of the PVC pipe and fitting connections are joined using PVC cement. For the PVC plugs used with the bubble level, RJ45 adapter and bottom (threaded) board insertion plug, it is necessary to seal the connections to prevent moisture intrusion. These connections can be sealed with wrapped tape (vinyl electrical tape and self-adhering rubber tape) or silicone caulking (Figure 14) (Figure 15).

Figure 14 – Silicone caulk applied to end cap of magnetometer housing.

Figure 15 - Silicone caulk applied to top and connector ports of the magnetometer housing.

Bill of Materials (as of 13JUN2026)

Description	Qty.	Mfr.	Mfr. Part No.	Vendor	Notes
2 in. PVC Schedule 40-Plug SPG	2	Charlotte Pipe	PVC021181800HD	Home Depot	
2 in. PVC Schedule 40 S x S x S Tee	1	Charlotte Pipe	PVC024001600HD	Home Depot	
2 in. x 2 ft. PVC DWV Schedule 40 Pipe	1	Charlotte Pipe	PVC072000200HA	Home Depot	Cut to 6-inch length
1 1/2 in. x 2 in. PVC DWV Hub x Hub Increaser/Reducer Coupling	1	Charlotte Pipe	PVC001020600HD	Home Depot	
1-1/2 in. x 2 ft. PVC DWV Schedule 40 Pipe	1	Charlotte Pipe	PVC071120200HA	Home Depot	
1-1/2 in. PVC Schedule 40 Female S x FPT Adapter	1	Charlotte Pipe	PVC021011400HD	Home Depot	
1-1/2 in. PVC Schedule 40-Plug MPT	1	Charlotte Pipe	PVC021131400HD	Home Depot	
CPLR C6A SHLD RJ45 IP68 PNL MT	1	L-Com	CG-C6A-FP	L-com	
Monoprice Cat6A 3ft Black Patch Cable Double Shielded (S/FTP) 26AWG 10G Pure Bare Copper Snagless RJ45 Fullboot Series Ethernet Cable	1	Monoprice	124335	Amazon	
Circular Bubble Level with Double Sided Adhesive Bottom Spirit Levels Leveling Measuring Tools (50x10mm), F-color	1	SIZIKJXGHWYI	J65	Amazon	Mounted using double-sided adhesive tape
White EVA Foam Sheet, 4mm Thick Foam Sheet for Cosplay, 59"x13.9" with High Density 86kg/m3 Eva Foam Roll for Art Craft Costume DIY Project (4mm White)	1	Do^{2}ping		Amazon	Many equivalent closed-cell foam products available
Silicone Caulk, 10.1oz., Window and Door Sealant, White	1	GE	2811093	Home Depot	Requires caulk gun. Alternate products are readily available.

Software Testing

The Raspberry Pi software is installed from the rm3100-runMag GitHub repository. The installation process is not described in this document.

To conduct a functional test of the magnetometer, the command line shown in the latest GitHub readme section was entered at the command prompt:

./runMag -b 1 -j -R 1f -r -H -M 23 -c 400 S <callsign or node id>

In Figure 16, the local grid square was temporarily used in place of the callsign or node ID.

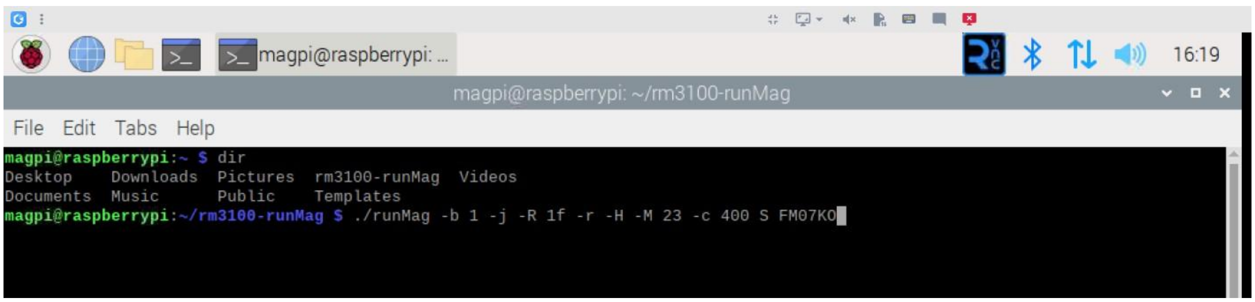


Figure 16 - Entering the command line data to start runMag. Here, the grid square was used as the node ID parameter.

Once the command line has been executed, the program will display the temperature reading and XYZ values from the magnetometer in one-second intervals. An example of the displayed data is shown in Figure 17.

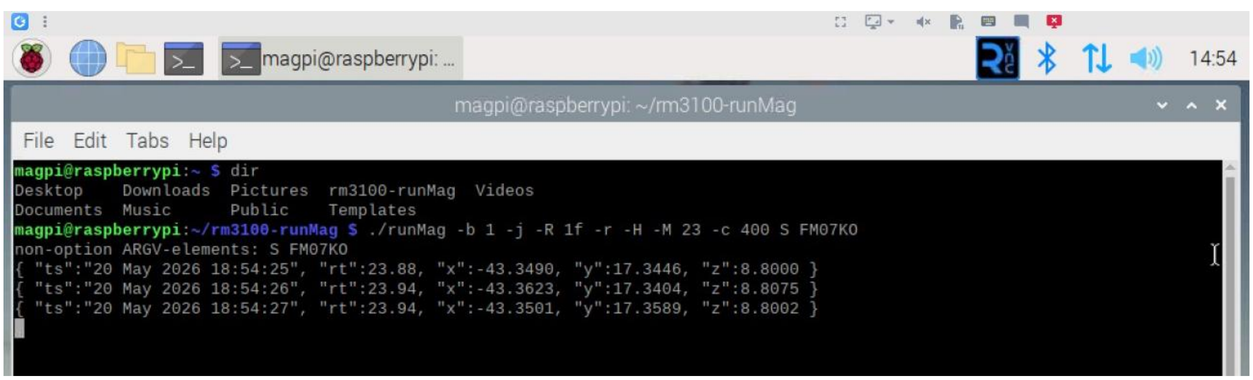


Figure 17 - Example data output once runMag is operational. The data is updated at one-second intervals. "rt" refers to the remote temperature sensor. "x", "y" and "z" refer to the RM3100 magnetometer board output values.

To stop program execution, type Ctrl-C.

Site Installation

Comment

Note: the cable assemblies and magnetometer installation procedure described in this section were specific to the author's location. Your installation needs may vary. Select the installation methods best suited for your location.

Cable Assemblies

Pre-made CAT6 cable assemblies were used during installation to avoid the need to purchase connector crimping tools. A 100ft run of CAT6 cable was installed inside a buried 4" corrugated drainage pipe along with coaxial cables used for other antennas. The magnetometer was located approximately 25ft away from the exit point of the pipe, requiring the installation of a direct-burial CAT6 extension cable. An IP68-rated female-female coupler was used to join the two cables (Figure 18). Although the coupler has an IP68 rating, an additional wrapping of vinyl electrical tape was used to provide further moisture protection (Figure 19). The coupler was then buried in the ground along the trenched cable path.

Figure 18 – Assembled IP68-rated female-female coupler.

Figure 19 - Vinyl electrical tape wrapping to provide further moisture protection.

Magnetometer Alignment

The magnetometer was installed in a hole approximately 2ft deep. The hole was dug using a digging bar and a post-hole digger. The magnetometer was placed in the hole and backfilled approximately halfway with loose soil to keep the unit upright (Figure 20). A stick was used to gently compact the soil. The bubble level on top of the unit was used to ensure that vertical orientation was maintained. The compass app on an iPhone (set to the magnetic compass reading), was used to roughly orient the magnetometer toward magnetic North.

Figure 20 - Initial installation of the magnetometer. The hole was backfilled halfway with loose soil to support the unit during final orientation.

Final alignment of the magnetometer required rotation of the housing, orienting the H and E axes based on the local magnetic field. To perform this field alignment step, the Raspberry Pi was collocated with the magnetometer (through a 10ft CAT6 cable) and operated in a "headless" configuration. The Raspberry Pi continuously polled the remote sensor board inside the magnetometer, providing readings corresponding to the orientation. A virtual connection was established between a laptop computer and the Raspberry Pi using RealVNC remote access software. Since newer versions of RealVNC require an internet cloud connection, connectivity was provided through a Mikrotik SXTsq5 wireless access point directed toward a home WiFi router inside the station location. A 12V battery was used to power the field equipment and a 12V to 5V DC-DC converter was used to power the Raspberry Pi. A network switch was used to interconnect the laptop computer, Raspberry Pi and access point. This configuration is shown in Figure 21 and Figure 22.

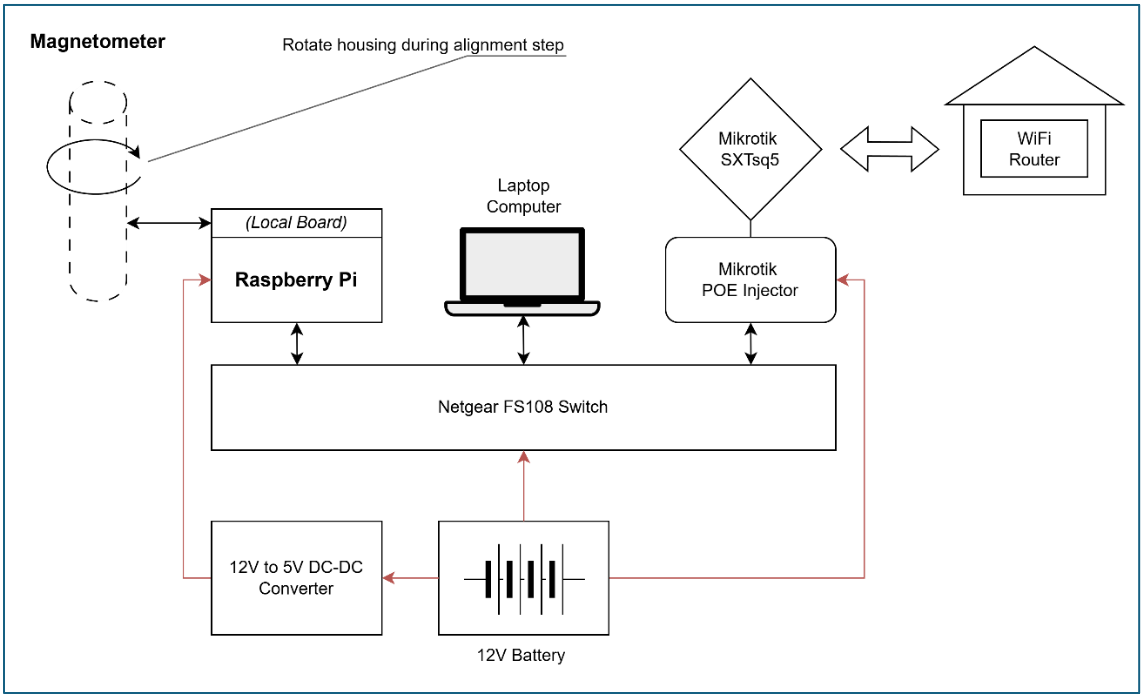
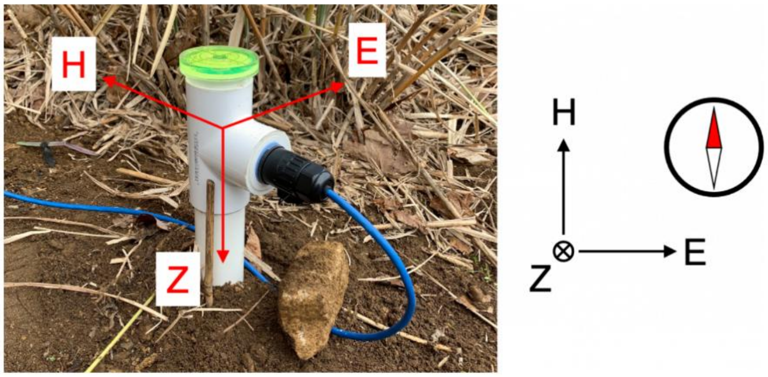


Figure 21 - Magnetometer field equipment configuration. The magnetometer was rotated while observing the data displayed on the laptop computer.

Figure 22 – Field equipment configuration used during the magnetometer orientation step. A 10ft CAT6 cable connected the magnetometer to the Raspberry Pi (inside the cardboard box). The laptop computer, Raspberry Pi and the wireless access point were connected using a network hub. A 12V battery (inside the green box) and 12V to 5V DC-DC converter were used to power the equipment.

Orientation Process

Although the magnetometer is oriented in accordance with the HEZ coordinate system, the runMag software displays the outputs of the X, Y and Z sensors on the RM3100 breakout board. The relationship between these vectors and the XYZ columns in the displayed data are shown in Figure 23.



Parameter Relationships				
HEZ Geomagnetic Coordinate Vectors		$\vec{Z}$	$\vec{E}$	$\vec{H}$
RM3100 Sensor XYZ Coordinates		X	Y	$-Z$
runMag Output Display Parameters (below)				
"ts"	"rt"	"x"	"y"	"z"
Orientation Goals			Minimize "y"	Maximize "z"

Figure 23 - Relationship between the various parameters associated with the magnetometer and the displayed data. Orientation is optimized when the "y" values are minimized and the "z" values are maximized.

Orientation requires <u>maximizing the "z" values</u> while <u>minimizing the "y" values</u>. Since the null response from the E vector is easier to discern during the rotation process, observe the "y"-column data while slowly rotating the magnetometer. During rotation, ensure that the unit remains level (per the bubble level indicator on top). Once the lowest absolute value is detected in the "y"-column data, gently compact the soil with a stick. Depending on the type of soil present, pouring about a gallon of water into the hole will make the mud "sticky," further helping to keep the unit stable during the rotation process.

Once the magnetometer has been oriented and no standing water is present in the hole, scoop the remaining soil into the hole by hand. Gently press the soil with your hands to compact it, but not so hard as to cause the unit to shift from its vertical position. After several days, the soil will again return to its naturally firm state.

Vegetation Control

Garden weed block fabric was placed around the magnetometer to prevent vegetation growth. A 3ft x 3ft sheet was cut, and an x-cut hole (creating loose flaps) was snipped in the middle of the fabric. The fabric was then placed over the unit and the CAT6 cable connected.

A rubber patio paver was modified to help keep the fabric in place and to further prevent vegetation growth. A spray can (roughly 2-3/4" diameter) was used as a template for the center hole cutout. The hole was cut using an oscillating multi-tool fitted with a narrow blade. To allow the paver to slip into position around the magnetometer housing, a slot was cut from one edge toward the center hole. This is shown in Figure 24 below.

Figure 24 – The rubber patio paver was modified by cutting a hole in the center and creating a slot on one side.

After installation, the paver was surrounded by 4” x 8” patio bricks. This was done to protect the paver from being damaged by garden tools such as string trimmers. Where the CAT6 cable came up from the ground, a small sheet of landscape fabric was cut and draped around the cable. 4” x 8” bricks were again used to hold the fabric in place and protect the cable. Once everything was in place, the excess fabric was trimmed using scissors.

To delineate the sensor location and prevent damage during lawn care, small wooden fencing was placed around the site. Each fence panel was 3 ft in length. A road reflector was also installed for added visibility. This configuration is shown in Figure 25.

Figure 25 - The magnetometer installed and properly oriented. The rubber paver and bricks cover the weed block material. The fencing and road reflector increase visibility and prevent accidental damage during lawn care.